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B.Tech Dissertation

**KNOWLEDGE-GRAPH ASSISTED RETRIEVAL
AUGMENTED GENERATION FRAMEWORK FOR
ANALYSIS AND ASSESSMENT OF TERMS OF
SERVICE DOCUMENTS**

Submitted in partial fulfillment of the requirements for the degree of

BACHELOR OF TECHNOLOGY

in

**Computer Science and Engineering - Artificial
Intelligence and Machine Learning**

by

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This is to certify that **Ansul Kumar** has successfully completed the project work entitled “**Knowledge-Graph Assisted Retrieval Augmented Generation Framework for Analysis and Assessment of Terms of Service Documents**” in partial fulfillment for the award of **Bachelor of Technology in Computer Science and Engineering - Artificial Intelligence and Machine Learning** during the year **2025-2026**.

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Declaration

We, hereby declare that the project titled “**Knowledge-Graph Assisted Retrieval Augmented Generation Framework for Analysis and Assessment of Terms of Service Documents**” is a record of original project work undertaken for the award of the degree of **Bachelor of Technology in AI and Data Science Engineering**. We have completed this study under the supervision of **Dr. Janani V S**, AI and Data Science Engineering .

We also declare that this project report has not been submitted for the award of any degree, diploma, associate ship, fellowship or other title anywhere else. It has not been sent for any publication or presentation purpose.

We have executed this project with code of research conduct as prescribed by the university.

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Abstract

The proliferation of digital services has led to users often validating Terms of Service (ToS) and privacy agreements that are lengthy, complicated and hard to understand in their entirety without legal skills. As a result, users may voluntarily agree to clauses that affect their privacy, financial responsibilities, and legal rights. In order to meet this challenge, this project introduces RiskWise, which introduces a web-based intelligent system that can automatically analyze ToS documents and also emphasizes potentially risky contractual clauses in an easy to comprehend and explainable way. The proposed architecture combines Natural Language Processing, Knowledge Graph generation and Retrieval-Augmented Generation in a single hybrid model.

Uploaded documents are semantically parsed and chunked and then embedded into document embedding in a pattern and similarity-based retrieval. Concurrently, structured subject-predicate-object relationships are taken to build a knowledge graph that reflects contractual entities and obligations.

A hybrid retrieval framework built upon vector similarity search and graph traversal extracts enriched contextual information, which is subsequently analyzed by a generative language model in order to detect risky clauses, grade them for severity and generate user-friendly explanations. The functional prototype was created with a modular full stack model using a React frontend, FastAPI backend, and graph database that supports relational and vector versions.

Case-based experimental testing validated the capability of the system to retrieve contextually relevant clauses, justify their nature, and allow interactive user exploration via conversational querying.

These findings show that hybrid Graph-RAG architectures can indeed be applied to facilitate contextual completeness and interpretability of automated legal document analysis. RiskWise bridges these gaps by translating complex contractual content into actionable insights, helping provide transparency for the digital experience, supporting informed consent, and educating the audience of users on what they can do in a digital landscape.

Keywords: Terms of Service Analysis, Knowledge Graph, Retrieval-Augmented Generation, Legal NLP, Hybrid Retrieval, Explainable AI

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GLOSSARY

Item	Description
AI	Artificial Intelligence; simulated intelligent behavior.
API	Application Programming Interface; rules for software communication.
DB	DataBase; structured data storage system.
Document Ingestion	Accept and extract content from uploaded documents.
Explainable AI	Interpretable model outputs; transparent reasoning and decision-making.
Graph Traversal	Walk relationships and connections to retrieve information.
Hybrid Retrieval	Combine vector similarity search with knowledge graph traversal.
KG	Knowledge Graph; structured representation of entities and relationships.
Knowledge Graph	Entity-relation graph; represents connections between concepts and facts.
LLM	Large Language Model; large neural network for text understanding.
NLP	Natural Language Processing; automatic language analysis methods.
RAG	Retrieval-Augmented Generation; retrieve relevant context before generating.
Risk Scoring	Numeric severity assessment for potential contractual risks.
Semantic Chunking	Divide text into meaningful segments based on semantic boundaries.
Semantic Embedding	Vector representation capturing meaning and semantic relationships.
ToS	Terms of Service; contractual rules and service agreements.
Triple Extraction	Extract subject-predicate-object facts from text.

Chapter 1

INTRODUCTION

Digital services have been growing exponentially and changed the way people are approaching technologies, allowing for the easy access to any apps, platforms, and the whole digital ecosystem. The expansion mentioned has also led to a lot of dependence on digital contractual agreements, such as Terms of Service (ToS), Privacy Policies, and End-User License Agreements (EULAs) which users have to accept prior to accessing services.

Although these agreements are important since they provide legal relationships between users and service providers, they are often very long and complicated. Because of this, the user finds it difficult to read and fully understand them before accepting the service, which raises concerns about whether or not the users are truly given informed consent and if transparency exists. But on the bright side, the recent development of AI and Natural Language Processing have made it possible to automatically analyse complex text without requiring prior knowledge of the content. Techniques like semantic embeddings, knowledge graphs, and Retrieval-Augmented Generation allow intelligent systems to extract useful insights from unstructured documents, while also preserving the meaning.

By applying the advancements mentioned above, within an AI-based framework to analyze legal agreements, the RiskWise initiative aims to generate intelligent and risk-aware insights, that help users better understand legal and other publicly available documents. This chapter discusses the background, motivation, and problem formulation of the proposed system.

1.1 Context and Background Study

As the extent of digital services and online platforms is expanding at a rapid rate, users are becoming more and more bound by Terms of Service (ToS), Privacy Policies and End User License Agreements (EULAs) in order to gain access to apps/services. What these documents do, is that they serve as legally binding agreements between service providers and users outlining their right, responsibilities, limitations of liability, data usage policies, termination conditions and other contractual terms related to the use of service. Such agreements are necessary in order to make legal relationships between parties in digital ecosystems. But since the general user experience is often poor, their practical effectiveness is often reduced. Although these agreements are important, studies and user behaviour data show that most users either rarely read these documents completely or do not fully understand them. The documents being too lengthy, having complicated legal language, etc. are the main reasons that contribute to this.

This general disengagement results in a great divide between user informed consent and user understanding, that has implications for the legitimacy of informed consent in the digital realm. Therefore, users could often agree to the terms and not even be aware of the possibilities for the complete collection of their data, unilateral changes to policies, subscription renewals, etc. This could lead to tremendous risks for their privacy, financial obligations, and access to services. New developments in Artificial Intelligence, specifically Large Language Models (LLMs) and Natural Language Processing (NLP), have proven tremendous scope for the automatic understanding of complex texts. On top of that, new techniques such as Retrieval-Augmented Generation (RAG), semantic embeddings, and knowledge graph representations enable structured understanding from unstructured texts (for example, semantic meaning and relationship). These new developments are useful when it comes to creation of intelligent systems that can assist in the comprehension of legal documents.

The RiskWise framework, in this technological context, is presented here as an AI-driven solution to deal with legal documents, identify potentially risky clauses and present findings in a simplified and user-centric way. Using the hybrid form of 2 retrieval and reasoning mechanism, the idea is to create a link between the complexity of a contract and user understanding to guide decision making in digital interactions.

1.2 Need Analysis

A lot of practical, technical and societal reasons for the need of an automated Terms and Conditions analysis system exist. First of all, modern digital ecosystems involve high levels of communication with a variety of online services, each one of which requires users to accept contractual agreements. Time-consuming and requiring legal literacy that non-standard users lack, reading these items by hand results in surface acceptance but not much consideration of the content.

Second, entities often include clauses granting wide discretion such as the exclusive power to change clauses, the broad flexibility to share data and the rights to cancel accounts. When no analytical support is provided, users are exposed to unknowingly accepting unsatisfactory terms within a contract that may threaten their privacy, financial obligations to customers or their access to services.

Thirdly, today's automated applications are primarily focused upon providing a description of the work that the users have done and fail to interpret the risk they have detected. Summarization may be an instrument to reduce the difficulty of documents but does not necessarily check for clauses, introducing foreseeable potential risks (and why they might be associated). Consequently, users need systems which can not only minimize but also strategically identify, classify and situate contractual risks (and why they might be associated).

Consequently, users need systems which can not only minimize but also strategically identify, classify and situate contractual risks. Furthermore, the widespread global trend towards digital privacy and protection of consumers as well as compliance requirements demands that we find mechanisms that allow us to promote transparency to give the right of informed consent to participants. It allows the system to identify material clauses, categorize potential risks, and walks the user through an interactive exploration of contractual content, and greatly increases their knowledge and decision-making capacity. Thus, there is a great demand for the proposed intelligent, user-friendly and privacy-preserving system for the assistance of users to weigh legal contracts prior to acceptance.

1.3 Problem Identification

The most important problem the project addresses is that users find it difficult to understand and assess complex legal documents in the digital service industry. This is due to several flaws in the system. Firstly, legal documents contain specialised legal language and complex linguistic structures that are difficult to understand for laymen. Secondly, the nature of agreements is such that it discourages complete reading, where selective attention (or complete avoidance) takes place. Thirdly, a layman may not have the necessary legal knowledge to understand the implications of the contract such that they are unable to point out clauses that might have a potential impact on their rights and obligations.

In addition, there are no easy to use tools for pointing out unfair or risky contract terms. The current state has low levels of interactivity, which might be detrimental to the users, as they may not be able to pose questions about certain contract terms. All these issues make the users not aware of the context, which makes them vulnerable to privacy infringements, financial risks, and service termination with too little understanding.

1.4 Problem Formulation

Building on the difficulties we have described here, this project explains the difficulty in terms of designing and putting into practice an intelligent system for automatically evaluating Terms and Conditions documents and giving users easily understandable, risk-aware insights. The suggested system must fulfil a number of usability and technical requirements in order to fully accomplish this goal.

Working with documents in different formats, extracting useful semantic information from legal text, identifying and classifying clauses based on potential risk, and offering explainable reasoning to support risk assessments are some of the essential system requirements. To facilitate deeper user understanding, the system must also allow interactive querying and offer a trade-off between user privacy requirements and performance.

This problem was chosen because of its applicability, interdisciplinary characteristics, and possible implications for society. The objective of this project is to

enhance user participation in digital decision-making processes and eliminate the complexity hurdle that comes with legal documents using modern NLP approaches, knowledge graphs, and interaction mechanisms.

Chapter 2

RESEARCH METHODOLOGY

2.1 Research Approach

We use a hybrid research design combining the principles of design science research and applied research methodology in implementing this study. The main purpose of the work is to design, develop, and evaluate an intelligent program which analyzes Terms of Service (ToS) documents and helps detect potentially risky clauses. The research of design science is evidenced by an operative artifact, which can be known as RiskWise, that uses the latest techniques of Natural Language Processing (NLP), Knowledge Graphs, and Retrieval-Augmented Generation (RAG) for solving a real-world problem. Simultaneously, we see the elements of applied research, as the project aspires for a practical technological solution to the challenge of user comprehension of complex legal agreements. Research involved iterative problem understanding, literature exploration, system design, implementation, and validation through both experimental and case-based testing. This allowed for iterative improvements in system architecture and analysis pipeline based on insights obtained during development.

2.2 Preliminary Research and Literature Exploration

We did a preliminary literature review to get an idea of the current research on things like AI-assisted contract understanding, unfair clause detection, and legal

document analysis. We acquired pertinent research articles and technical materials from academic databases and repositories (e.g., Google Scholar, arXiv, IEEE Xplore) via reference tracing from specific publications. The literature search focused on keywords such as Terms of Service clause detection, unfair ToS detection utilising NLP, legal document analysis, knowledge graph-based legal reasoning, Retrieval-Augmented Generation for legal documents, and Graph-RAG architectures. We looked at about 10 to 15 papers and chose 6 to 8 to read closely because they were new, relevant to legal document processing, methodologically innovative, and useful for explainable AI-driven legal analysis.

This exploratory phase helped us find important research areas that need more study, such as the fact that traditional NLP models are not good at structural reasoning, there are no clear ways to identify risks, and there is a need for hybrid search strategies that combine semantic similarity with relational reasoning.

2.3 System Development Methodology

The RiskWise system was made using a method that involved making prototypes and trying things out over and over. The project goes through a number of iterations. It starts with exploratory exercises using different clause detection models and ends with the design of a hybrid architecture that includes building a knowledge graph and RAG-based reasoning.

First, we did comparative analyses of transformer-based models for clause extraction tasks. This helped us understand the pros and cons of text-based methods. Based on these results, the system architecture was gradually improved to include structured ways to store legal information and ways to find it that take into account the context. The iterative method made the system’s parts modular, such as the document ingestion, semantic chunking, embedding generation, knowledge graph construction, hybrid retrieval logic, and generative reasoning modules. Every release of the work gradually improved the functionality of the features without changing how they worked with earlier versions. This way, the system’s abilities can gradually improve.

2.4 Proposed Methodological Framework

The methodological framework of the project is based on a multi-stage processing pipeline designed to transform unstructured legal documents into an interpretable knowledge representation and risk analysis output. The major methodological stages are summarized below.

2.5 Document Ingestion and Semantic Parsing

It starts when users upload ToS documents in supported formats. Text extraction methods are used to extract raw textual content from documents, and semantic segmentation is performed into meaningful chunks. Semantic chunking, as opposed to fixed-length segmentation that discards contextual boundaries such as clauses and sections, can enhance the quality of downstream retrieval.

2.5.1 Embedding Generation and Vector Indexing

Semantic embeddings are obtained for each segment of the text chunk, based on our domain-adapted embedding models which can represent legal semantics. Similarity-based retrieval is performed by storing such embeddings in a vector index, which facilitates efficient similarity-based retrieval during query processing. The vector component of the system is responsible for fuzzy matching and semantic equivalence detection of legal terminology variations.

2.5.2 Knowledge Graph Construction

Alongside embedding generation, the system builds out a knowledge graph for the document. Large Language Models are used to derive subject-predicate-object triples that describe relationships between entities such as service providers, users, data, and obligations. These triples are mapped to a structured legal ontology and stored in a graph database, preserving relational and logical dependencies within the document.

2.5.3 Hybrid Retrieval Mechanism

In the system we implement a hybrid retrieval strategy involving vector similarity and knowledge graph traversal. Vector retrieval retrieves semantically relevant clauses, and graph traversal finds relationally connected information which don't share the common vocabulary. Incorporating these retrieval signals forms the comprehensive background information that further justifies the reasoning ability.

2.5.4 Generative Reasoning and Risk Analysis

A generative language model set up as a legal risk assessor gets the combined contextual information. The model analyses clauses, finds potentially risky parts of contracts, gives them severity scores, and makes natural language summaries that are easy to understand based on the context it finds. Using citations to prompt further increases the transparency and traceability of outputs.

2.6 Evaluation Strategy

The proposed system underwent a system evaluation utilising qualitative case-study analysis. We looked at a few Terms of Service documents with the developed pipeline to see how it works, how well it can find risks, and how well it can respond. Experimental phases culminating in comparative model evaluation have yielded additional empirical evidence for baseline clause extraction performance. A manual inspection of system outputs was conducted to confirm the relevance of identified clauses, the appropriateness of risk categorization, and contextual accuracy of generated explanations. Since the project was exploratory and prototype-based, it was thought that a qualitative validation method would be best.

2.7 Validation and Reliability Measures

To ensure the system's results are reliable and trustworthy, a method has been set up that includes several validation steps. Retrieval-Augmented Generation allowed

the model to build its answers based on a document it had found instead of just using the knowledge it had already learned. Knowledge graph grounding made sure that clauses were consistent with each other, which made reasoning even harder. Other ways to check reliability included structured parsing of the information that was taken out, fallback mechanisms for dealing with model outputs that weren't consistent, and using supporting text segments in context. This combined method lowers the chance of hallucination and also makes it easier to understand the results of system-generated analysis.

Chapter 3

LITERATURE SURVEY AND OBJECTIVES

3.1 Critical Review of Literature

A growing prevalence of digital services has resulted in a need for Terms of Service (ToS) agreements governing user-provider relations. However, the complexity and length of such documents have given rise to a number of important research developments in the automated analysis of the contractual content itself. The paper extends the research on legal infomatics, Natural Language Processing (NLP), and knowledge graph-based reasoning into the domain of interdisciplinary issues. Among all the research, the CLAUDETTE framework is arguably the most important and has introduced automated identification of potentially unfair clauses included in ToS agreements, employing ML algorithms.

The paper proved that it could offer better precision and recall in comparison to other classification techniques. Though it has come so far as a work of such significance to the field, the CLAUDETTE model was largely dependent on the classification of clauses and did not consider the contextual dependencies that could exist between clauses, i.e., it did not reason on complex structures of contracts. Research in a similar area to contract law has revealed the consequences on society in general in regards to unfair contract provisions and the importance of intervention in such matters. As for unfair contract term legislation, research in the legal area of unfair contracts revealed the discrepancy between service providers

and service consumers, emphasizing the need for instruments that would promote service provision and awareness among consumers.

Although the above works give a good theoretical foundation they do not suggest any technological tools that can be automated for consumer use. Recent studies on the interface of AI and contract law suggest neural and machine learning technology for the automation of interpretation of contracts. The above studies suggest that AI can help in identifying unfair terms in contracts and can assist legal practitioners in interpreting contracts. However, it also suggest that AI is limited in terms of interpretation, content and context.

Recent breakthroughs achieved using Retrieval-Augmented Generation (RAG) and graph-based techniques have created new opportunities for dealing with these problems. The use of Graph RAG has highlighted the importance of legal texts as structured texts that have hierarchical and relational relationships. The application of Graph RAG has provided better contextual retrieval compared to vector-based RAG methods, though with a focus on legislative texts rather than private contract agreements.

Benchmarking studies on measuring large language models for contract review have also identified gaps in thinking over long context and clause retrieval. Linguistic models may not identify information on large documents, and hence the interest in hybrid models. Hybrid architectures for applications such as financial analysis using Hybrid Graph-RAG models that include vector similarity search and graph traversal have also been proposed. While promising, such techniques were not specifically developed for the analysis of legal documents, such as contracts, or for identifying clauses based on risks. To summarise, the literature reviewed above clearly shows that there has been substantial improvement in the automated techniques for contract analysis, as well as the outstanding issues, which include structural reasoning, explainability, and user-centered risk interpretability. This has greatly affected the proposed RiskWise framework.

3.2 Literature Collection & Segregation

The studies evaluated in this study systematically identified academic repositories like Google Scholar, arXiv, IEEE Xplore, and reference tracing from relevant publications. The identified studies were grouped into thematic groups for analysis.

In the first category, studies on the detection of unfair clauses and legal texts' classification, such as the CLAUDETTE framework, as well as other studies on machine learning, were identified.

The studies provided a basic understanding of risk identification through the application of automation 12 techniques, though most of the studies were limited to classification. There were also studies on legal and policy studies regarding unfair contract terms and consumer protection policies. The studies provided domain knowledge on different problematic clauses. The third category of studies included AI-based legal analytics and understanding of contracts, which discussed the development of neural language models. The fourth category of studies included those related to Retrieval-Augmented Generation, Knowledge Graph, and hybrid retrieval models that have come into the limelight in recent times.

This study has been a methodological inspiration for the integration of semantic and structural models in document understanding. This thematic categorization allowed the identification of complementary studies among research areas, which also helped in identifying methodological gaps.

3.3 Summary of Literature

According to a literature review, automated legal document analysis is progressively moving from traditional machine learning classification techniques to more sophisticated language model-based techniques. Though classification-based systems are geared towards recognizing particular clause patterns, there is a limit to their explanatory potential and a lack of consideration of clause interactions. Though providing few computational results, the legal literature highlights the need to address clauses of unfair contract terms from a social perspective.

Though AI-based systems have certain limitations related to context, they have a high potential for understanding text. As a measure to improve context aggregation, there is a focus on Graph RAG, and its potential is emphasized, including structural representation and multi-hop reasoning. However, these architectures have yet to undergo a thorough implementation in consumer-oriented ToS analysis. Therefore, in conclusion, the literature indicates the feasibility of ToS analysis and the requirement for systems comprising structural reasoning and semantic retrieval and risk assessment in an explainable manner.

3.4 Identification of Gaps

Based on the critical review and synthesis of the literature surveyed, some research gaps were identified that form the basis for the development of the proposed system.

Firstly, it has been observed that the existing systems for detecting unfair clauses, for the most part, rely on classification models that classify each clause individually. Though it may have been successful in identifying the clauses of a particular kind, it may not always be possible to take into consideration the context of the document in which these clauses may be presented. It may not always be possible to arrive at a reliable risk assessment depending on the relationship between clauses, supporting statements, and references, which may or may not be applicable.

Secondly, it may not always be very transparent with regard to the results obtained by existing AI-based contract analysis tools. Although these tools are successful in identifying clauses that may be potentially problematic, they are often not very forthcoming about the rationale used for risk classification.

Third, the conventional Retrieval-Augmented Generation models generally work directly with flat forms of texts. This may lead to the problem of the "fragmentation of contextual information," where "semantically related clauses that are spread out in different parts of the document may not be retrieved together." As such, "crucial cross-referenced information may not be taken into account in the reasoning process."

Fourthly, it has been shown that there is a limit in the way in which large scale models of language process complex relational dependencies in a document. In agreements where clauses and relationships are interdependent, it has been shown that there is a difficulty in maintaining a coherent state of understanding in large-scale models of language.

Finally, existing research on the domain of automated contract analysis tools has been found to be mostly focused on the solutions that are meant for legal professionals and enterprise applications. These applications have been found to be used by domain experts and may require output in technical or legal formats that may not be easily understandable by a non-expert. This shows that there is a problem

related to a lack of user-friendly tools that can interpret complex contract risks in an understandable format for a non-expert.

3.5 Need for Research/Project

The gaps identified above indicate that there is a noticeable need for a smart system, providing semantic and structural representations on ToS documents and explaining results to non-expert users. Such a system would be beneficial for increasing transparency in digital agreements, promoting informed consent, and increasing consumer awareness. Additionally, the increasing usage trend of generative AI can be leveraged to develop hybrid architectures targeting limitations of current techniques. This project seeks to bridge the gap between automated detection and human-understandable explanations of contractual risks by integrating knowledge graph reasoning with RAG-based language models.

3.6 Problem Statement

Despite the existence of Terms of Service contracts on digital platforms, users are not prepared to understand complex contractual terms and clauses that could negatively affect their rights, privacy, financial commitments, and service access terms.

The contracts are complex, lengthy, and use legal terminology that is not understood by any layman, which results in the acceptance of terms of the contract by the users without proper consideration. In terms of the difference between acceptance of users and awareness of users, this is one of the biggest risks in terms of decision-making in a digital ecosystem. In terms of the analysis of contracts, though a few automatic methods have been proposed, there are several such methods that have some serious flaws.

Some are largely based on clause-level classification methods that are unable to reason in context and fail to establish the relationship between distributed contractual clauses. Some methods provide simple and generic models of summarization based on language models that do not specifically point to potentially risky clauses or explain them in a manner that is interpretable to establish risks. Therefore,

the need for a system that can mediate with contextual awareness and structured reasoning as well as user-centered interpretation, is still required.

Hence the objective of the present paper is to develop an intelligent and transparent system that can analyse and comprehend the Terms of Service documents in an automated manner, as well as detect the critical clauses of the contract in a manner that can be easily interpreted by the user. The solution to the problem would be achieved through the development of a hybrid retrieval and reasoning paradigm to assist the use of semantic retrieval, knowledge graph construction, and generative language models.

3.7 Objectives

The main aims and objectives of the project are as follows:

1. Design and develop a hybrid retrieval system that combines vector similarity search and knowledge graph traversal for legal document analysis.
2. Design and develop an automated system that can extract structured knowledge representations from unstructured Terms of Service documents.
3. Design and develop a system that can identify and classify potentially risky contractual clauses using context-aware reasoning.
4. Design and develop a system that can provide explainable natural language interpretations of identified risks.
5. Design and develop an interactive web-based interface that can allow users to upload documents and perform automated risk analysis.

3.8 Limitations of the Study

The study has a number of limitations. It is based on language models, whose performance may depend on the capability of the model and the complexity of the documents. The system is mainly suited for English-language documents and is not applicable in multilingual settings. In addition, the study has used qualitative

methods of evaluation because the study is exploratory in nature, and large-scale quantitative benchmarking is not within the scope of the study. Resource constraints related to local model execution and the variability of LLM outputs are also limitations of the current system.

Chapter 4

ACTUAL WORK

4.1 Methodology for the Study

By implementing the RiskWise system based on a multi-stage process, an unstructured legal document like a Terms of Service is converted to a structured knowledge representation, which is accompanied by user-related risk insights.

Unlike in the research methodology as described in Chapter 2, which is more focused on explaining issues related to the conceptual design and research strategy, in this section, a description of the operational workflow developed in the course of creating the framework is presented. The operational workflow is designed to process raw legal texts and generate understandable output in a systematic manner through a set of analytical steps.

The process begins with a "document ingestion part" in which the user can upload documents containing "Terms of Service" in a web-based way. The proposed service aims to assess the documents uploaded by the user, as well as documents in multiple formats. The text extraction part of the documents will be performed, keeping the logical order of the documents.

Then, a semantic segmentation component is applied to text data in which a coherent chunking of text is performed at clause-level granularity. The mechanism for text segmentation is designed to maintain context and facilitate text retrieval and reasoning, which is a subsequent step. Subsequently, a semantic embeddings component is applied in which embeddings for each segmented text component are generated using embedding models designed for legal text and its subtle semantics.

These embeddings are vectors of textual meaning allowing similarity-oriented retrieval over semantically related clauses even beyond lexical overlap. The generated embeddings are stored in a vector indexing mechanism that allows fast and scalable nearest-neighbor search during analysis and query processing.

These embeddings are vectors of textual meaning allowing similarity-oriented retrieval over semantically related clauses even beyond lexical overlap. The generated embeddings are stored in a vector indexing mechanism that allows fast and scalable nearest-neighbor search during analysis and query processing.

Apart from using the model for the embedding of the generation, the language model can also be used for extracting subject-predicate-object triples from the text segments, which can form a basis for representing the knowledge graph. The representation of the knowledge graph can give an account of the relational dependency of the entities such as users, service providers, permissions, obligations, contractual actions. By connecting the components of the documents, the knowledge graph enables the relational analysis that can connect the context beyond the apparent textual similarity.

In addition with the aim of using the semantic knowledge as well as the knowledge of the structure, a new hybrid retrieval technique has been developed that can include vector similarity search as well as graph traversal. This will enable one to retrieve not only contextually relevant information but also relationally related information in a way that enriches the context of the information for a comprehensive analysis thereof. The hybrid-technique in information retrieval, as proposed above, will enable one to transcend the limitations of different information retrieval techniques and enrich the context of the information.

Lastly, the contextual information retrieved is processed by a generative reasoning module set up for risk-based processing. This module provides analyses of the context, and uses it to categorize potential risky clauses, to make severity judgments to help derive a score and categorize risk-types and to provide explainable summaries in natural language based on evidence retrieved. Combining retrieval grounding with generative reasoning, the system's outputs are interpretable, contextually aware, and consistent with document details.

The RiskWise system manages the efficient, explorable and context-sensitive analysis of contractual documents through the multi-step and coordinated approach,

making it possible to process intricate legal agreements into general users’ accessible results.

4.2 Experimental Work

The experimental stage of the project was characterized by iterative evaluation of different methods for automated clause detection and risk identification. The initial experimentation stages involved experimenting with the transformer models for question answering for clause extraction. This provided the initial insights for what these models can achieve for legal text understanding. Other experimentation stages involved validating different components of different models, including the accuracy of the semantic chunking schemes, embedding-based retrieval methods, and knowledge graphs.

ToS documents from a variety of digital platforms were examined and processed in the pipeline to analyze system robustness for different document structures and legal formulations using multiple ToS documents as case studies. Overall, the comparative analysis of the retrieval results confirms that hybrid retrieval mechanisms achieved greater contextual completeness than purely semantic retrieval.

In addition, testing the local and cloud-based inference modes facilitated a comparison in terms of performance trade-offs in terms of computational efficiency and response latency. Experiments were conducted to facilitate fine-tuning and validation of the proposed hybrid GraphRAG framework.

4.3 Analytical Work

We manually reviewed the retrieved clauses for semantic similarity search for contextually appropriate segments corresponding to user queries and analysis tasks. The extracted relationships were checked to be accurate in terms of logical consistency to ensure that the extracted relationships corresponded to that of identified entities.

The results obtained from graph traversal were validated to ensure that the contexts provided in the relational contexts gave additional information apart from

the results provided in the vector retrieval results. The results obtained from the risk analysis module were validated in terms of their understandability, accuracy and consistency with the document content.

The severity scores and the risk categories were also validated to ensure that they addressed the issues related to the interests, such as changes in contracts, liabilities and permissions related to data usage. Thus, the context grounding has been appropriately achieved for the interpretation of the user.

4.4 Modeling, Analysis & Design

A modular and multi-tiered architecture is used to represent the system. This system has its end-user interface layer, where there is the web interface and the presentation layer, the system that processes the provided information, the processing layer, and the storage system itself, the storage layer. The presentation layer has the role of designing a user interface that is not just intuitive but also friendly. The processing layer is where the intelligence of the system can be found. This system has six different modules in the processing layer of the system and these are the text extraction module, the semantic chunking module, the embeddings module, the knowledge graph module, the hybrid retrieval module and the generative reasoning module. All the modules in the system are independent of each other, though the objective of all the modules in the system is the same, at least in the processing layer of the system. This storage layer has the graph database itself.

Furthermore, this design aids in retrieval and reasoning by attaching semantic and structural information together. The key factors in how the system was designed are modularity, scalability and clarity. Because the system separates the vector and graph representation of information, different retrieval options can be utilised.

The modular processing components can accommodate future growth and enhancements to the processing models without the need to redesign the entire system.

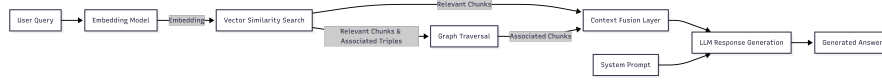


FIGURE 4.1: Retrieval Mechanism



FIGURE 4.2: Frontend Architecture

4.5 Prototype & Testing

To demonstrate end-to-end analysis functionality, the RiskWise system has been created as a prototype consisting of a web user interface integrated with back-end capabilities to process and analyze documents. Users can upload ToS documents, check their processing status, view risk results classified by type, and interact with the system using a conversational query interface. The prototype testing was performed by simulating various user scenarios (e.g. uploading a document, displaying risk analysis results, and answering queries in context). Functional testing was performed to verify that each stage of the pipeline was executed correctly and integration testing was conducted to verify that the front-end and back-end systems interacted correctly. For performance testing, processing time was measured for different document sizes and inference configurations to ensure there were expected differences in latency between local and cloud executions. Unsupported formats and invalid inputs were included in the testing to validate error handling and confirm strength of the system. The overall prototype testing indicated compliance with the functional correctness and usability requirements of the system.

4.6 Algorithms and Flowcharts

The operational workflow of the RiskWise system can be represented through algorithmic steps describing the document analysis pipeline.

Algorithm: RiskWise Document Analysis Pipeline

1. Start

2. Accept document upload from user
3. Validate file format and generate document identifier
4. Extract textual content from document
5. Perform semantic segmentation into text chunks
6. Generate embeddings for each chunk
7. Store embeddings within vector index
8. Extract knowledge triples from chunks
9. Construct knowledge graph representation
10. Perform hybrid retrieval combining vector similarity and graph traversal
11. Provide retrieved context to generative reasoning module
12. Identify risky clauses and assign severity scores
13. Generate explainable summaries and present results
14. Enable interactive user querying
15. End

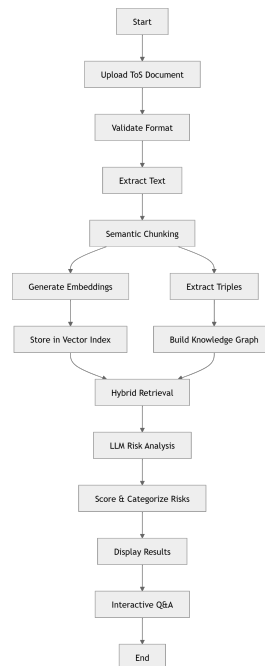


FIGURE 4.3: Document Analysis

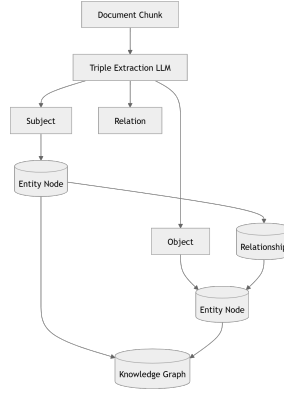


FIGURE 4.4: Knowledge Graph Generation

4.7 Codes / Standards

RiskWise system was developed in accordance with recognized standards of software engineering, with established tools in data handling, industry-oriented practices and standard methods to maintain reliability, integrity, maintainability, compatibility and responsible behavior. Due to being a work in the automated legal report inspection area, transparency and modularity and explainability at all stages of development were the focus of the code.

Standards for Software Development

The framework was mostly developed in Python code and was written in a standard coding fashion to make it easier to read, collaborate, and maintain. To make sure that the code was readable, standard naming and modular programming practices, as well as organizing modules in an organized way, were followed. The main functional modules, such as document processing, embedding generation, knowledge graph construction, retrieval logic, and reasoning modules, were developed as separate modules.

API and Integration Standards

Both front-end-side and backend layer functionality of the interaction between pieces of system programming was handled by creating code on, for example,

RESTful API principles of the RESTful API framework to enable communication between API components. Data exchange was done through standard HTTP request-response mechanisms, and JSON format which the data was used as the main data format because of its lightweight layout and the ability to use this media type for its lightweight and applicable platform. These integration methods were leveraged to facilitate easy integration of document-based service-ingestion systems, retrieval pipelines, retrieval methods, and interfaces in the user interface components were optimized by seamless orchestration.

Data Management and Data Protection Standards

Since user uploaded documents could include potentially proprietary contractual information, privacy-aware data handling practices were accounted for in the system development. We used uploaded documents as workstopped files and did not keep superfluous metadata and used temporary caching mechanisms for each iteration of the same document to enhance performance. The approach is focused on minimizing data and treating analysis artifacts with control; both data protection principles and ethical AI development principles are well-accepted.

Ethics in AI and Responsible Usage Guidelines

The RiskWise system was developed with a focus on responsible AI use. The returns on risk analysis are structured in a way that is based on the document context retrieved to mitigate the issue of hallucination in terms of how the generated explanations can be made more transparent in terms of interpretation. The system has a purposeful approach to its usage in the promotion of user awareness rather than legal advice and the results produced in the form of informative insights.

NLP Standards and Knowledge Representation Standards

Normalization methods in NLP, such as text normalization, sentence segmentation, and semantic chunking, are performed on a periodic basis for data preparation in NLP. The embedding methods used were based on the transformer modeling paradigm that has been popular in modern semantic retrieval models. Additionally,

the development of the knowledge graph was based on the subject predicate object representation paradigm that has been widely used in knowledge engineering, enabling data structure relationships in all document materials.

System Design and Documentation Standards

Project documentation is guided by academic and engineering documentation conventions for a modular system description, architectural diagrams, algorithmic representation, experimental verification and validation. We included flowcharts and system diagram of system functionality, which will facilitate reproducible and technically transparent design, so that the system design can be understood, reviewed and extended by any future researcher or developer. Following these coding, integration and operational conventions, RiskWise system ensures enhanced reliability, scalability and interpretability while ensuring the responsible processing of user provided contractual data. These behaviours all facilitate the development of a scalable and resilient platform for in the automatic interpretation of legal documents.

Chapter 5

RESULTS, DISCUSSIONS AND CONCLUSIONS

5.1 Results & Analysis

A series of case-based experiments based on Terms of Service documents gathered from various online platforms were used to assess the RiskWise system. Using its hybrid pipeline, the system processed the uploaded documents to create knowledge graphs, produce semantic representations, and provide risk-informed insights based on the contextual evidence gathered. Based on observations from the system implementation and experimentation as well as the synthesis of the literature, the results are discussed.

5.1.1 Findings of Literature Survey

The literature survey conducted as a part of this study has been very informative in understanding the status of automated contract analysis and research in legal AI. The following points have been raised based on the studies surveyed:

- The current state of unfair clause detection is based on classification models analyzing clauses independently.
- Most AI-based contract analysis tools lack contextual models that understand the relationships between clauses.

- Lack of explainability is a problem, and most tools are able to detect risks but lack the ability to explain their reasoning.
- The traditional Retrieval-Augmented Generation model analyzes text in a flat structure, which can lead to the loss of contextual information.
- Most tools are designed for legal professionals and not end-users.

Analysis: The literature survey conducted emphasizes the need for a system that bridges the gap between automated clause detection and user-friendly explainable analysis. The RiskWise framework fills this gap by incorporating hybrid retrieval based on semantic similarity and relational reasoning.

5.1.2 Results of System Implementation

Contextual Retrieval Performance

- The segmentation of the uploaded documents was done into semantically significant chunks.
- Vector space representation made it easier to retrieve similar clauses based on similarity.
- Traversal of the knowledge graph helped to expose contractually relevant information.
- Hybrid retrieval combined the results to create contextual evidence.

Result: Improved contextual completeness and efficient retrieval of distributed contractual provisions.

Risk Detection and Explanation Quality

The ToS risk identification system was tested using example ToS documents to determine the functionality of the system in identifying risks associated with typical ToS documents.

- Unilateral modification clause identification

- Liability limitation clause identification
- Data usage and privacy permissions identification
- Termination and service restriction clause extraction

The explanations provided were tested for their relevance and interpretability. Result: Risky clauses were successfully extracted along with relevant explanations.

Interactive Querying Performance

The conversational interface was tested to determine the capabilities of user interaction and exploration of context.

- Users were able to ask follow-up questions about risks identified.
- Contextual evidence retrieved supported informed response.
- Responses were still relevant to document content despite turns in interaction.

Result: Successful exploration of contractual risks using the conversational interface while still grounded in context.

5.1.3 Overall Observation

The combined results demonstrate that the RiskWise system successfully transforms complex legal agreements into structured representations and interpretable insights. The hybrid Graph-RAG pipeline enabled context-aware retrieval and explainable reasoning, validating the system’s suitability for assisting users in understanding contractual risks associated with digital services.

5.2 Comparative Study

This study compares a traditional approach for clause extraction, text only; and a newly proposed hybrid Graph-Recurrent Attention to Graph (RAG) Model for

clause extraction. The existing literature demonstrates that transformer based question-answering models exhibit reasonable efficiency for clause extraction but fall short of contextual completeness and relationship oriented reasoning. Conversely, the proposed hybrid model in the RiskWise framework performed better with respect to capitalising upon contextual context, and demonstrating relationship between clauses. In addition, the knowledge graph aspect of the model facilitated identification of obligations/permissions which related but were not derived from semantic similarity. Also, the hybrid model provides contextually appropriate citation for both the text segments derived from clauses and the evidence for the relationships, making them contextually interpretable as well. Ultimately, this comparative analysis demonstrates that combining structural representations of contracts with semantic extraction improves reliability and interpretability in automated contract analysis systems.

5.3 Discussions

The proposed project clearly demonstrates the applicability of the knowledge graph and Retrieval-Augmented Generation paradigm to be combined for addressing legal document understanding tasks. By combining semantic as well as relational approaches, the system bridges the gap that has been noticed in the past. It is also important to note that the structural representation helps in improving the contextual completeness, especially when related clauses are identified in different spots. Additionally, the conversational interaction framework helps in improving accessibility by allowing users to inquire about document understanding in natural language queries. On the other hand, the study also points out some intrinsic limitations of generative AI systems, such as variations in output and reliance on model capability. Other trade-offs in performance between local and cloud inference modes have also been noticed, which are indicative of real-world deployment aspects.

5.4 Project Cost Estimation

The RiskWise project was primarily developed using open-source technologies and existing computational resources, resulting in minimal direct financial expenditure.

However, a conceptual cost estimation is presented to reflect potential implementation and operational expenses.

TABLE 5.1: Project Execution Cost

Project Execution Cost			
No	Category	Description	Estimated Cost (Rs.)
1	Software Tools	Open Source Frameworks and Libraries	0
2	Hardware Resources	Personal Computing resources	0
3	Cloud Usage	API inference and hosting (estimated)	2000
4	Miscellaneous	Testing and internet usage	1000
Total Cost			3000

TABLE 5.2: Service Deployment Cost

Service Deployment Cost		
No	Component	Estimated Monthly Cost(Rs.)
1	Cloud Hosting	1500
2	Model Inference API	3000
3	Database Hosting	1000
Total Cost		5500

5.5 Project Impact

5.5.1 Environmental Impact

This application is based mainly on software and does not involve large amounts of physical resources. Cloud-based processing may increase indirect energy consumption. However, due to clever retrieval mechanisms and caching strategies for the best possible retrieval, there is less effort for computer overhead because redundant computation is avoided.

5.5.2 Societal Impact

The significance of the RiskWise project as a whole is on the side of humanity. The system encourages transparency and informed consent in digital environments.

The system assists users to make educated choices regarding digital service adoption through comprehension of terms, by reducing the complexity of contractual contracts. It helps make a dent in consumer knowledge, enhances digital literacy, and gives everyone equitable access to information.

5.5.3 Economical Impact

Economically, the system demonstrates potential to reduce legal consultation costs for preliminary contract understanding and risk identification. Organizations may also benefit from automated contract review capabilities, improving operational efficiency. The architecture’s reliance on open-source technologies further enhances cost-effectiveness and accessibility.

5.6 SDG Goal Compliance

The project satisfies a number of the United Nations Sustainable Development Goals (SDGs). The project satisfies SDG 9 (Industry, Innovation, and Infrastructure) through the development of innovative AI-driven infrastructure for automating document analysis. The project also satisfies SDG 16 (Peace, Justice, and Strong Institutions) through the promotion of transparency and fairness in contractual relationships and through the enhancement of access to understandable legal information. The project indirectly supports the promotion of inclusive digital ecosystems and responsible technological innovation through the empowerment of users with knowledge about digital agreements.

5.7 Conclusions

The current study has been able to achieve the objectives of designing and implementing a hybrid Graph-RAG approach for the automated analysis of Terms of Service. The system has been able to efficiently combine semantic embeddings, knowledge graphs, and generative reasoning power. The RiskWise system designed in the current research study demonstrates the feasibility of transforming complex legal texts into interpretable insights using context-aware analysis and reasoning.

The hybrid search strategy was able to enhance the completeness and correctness of context awareness over the baseline approaches, and the conversational interface was able to enhance accessibility for non-experts. The current research study introduces a new contribution to the legal AI community by offering a new consumer-oriented application of hybrid search architectures.

5.8 Scope for Future Work

RiskWise sets up a base framework for automated Terms of Service analysis based on hybrid retrieval and reasoning methodologies. Nevertheless, there are still many research and development opportunities to improve system capability and expand its flexibility as well. Immediate direction is the integration of multilingual processing to support analysis of contractual documents even in several languages.

The use of multilingual embedding models and language-agnostic large language models will ensure that the user base will be global in nature, and also expand the usability beyond English-language agreements. In addition, the incorporation of Optical Character Recognition (OCR) processing would also allow scanned and image-based documents to be processed and thus extending the coverage on document-based applications and its real life relevance.

Another major extension is the ability to visualise knowledge graph structures more graphically. Interactive and dynamic graph visualizations would enable users to explore the relationship and the implications of contractual entity vs obligations vs permissions that makes them more interpretable and can improve understanding in terms of how to understand the document semantics.

This visualization could also be helpful to legal practitioners in exploratory contract analysis and auditing. Personalization methods could also be researched under the model of Risk Analysis in the future. Customization of risk profiles would enable users to set priorities for different categories such as privacy, financial obligations, etc.

Adaptive analysis, taking into consideration the user preferences, may also prove helpful in improving the relevance of the context-aware interpretation of the risks. The idea of browser extensions appears to be promising in the context of ToS analysis in real time. This form of integration may prove helpful in enabling the

analysis of the agreements that are encountered in the online interaction in a proactive manner before agreeing to the language of the contract.

Also, enterprise-oriented batch processing capabilities could let companies review multiple contracts with various vendors and perform 32 comparative contract risk assessments. Methodologically there are opportunities to broaden the reasoning ability of the enterprise, by including legal terminology and databases for their domain knowledge.

For example, the inclusion of external legal standards and legal precedent data in the augmentation of the building of the knowledge graph could potentially allow for the development of semantic richness which could allow for the development of complex reasoning in compliance. The focus of the research agenda will also include the participation in the empirical validation of the proposed framework through the conduct of quantitative benchmarking against annotated legal datasets as well as user performance in relation to metrics of system performance in relation to the identification of clauses that can be achieved in relation to the retrieval of the same information. User studies can also be included in the identification of user performance in relation to usability, trust and interpretability.

The potential to explore and research possibilities for distributed infrastructures in data processing, optimally indexed storage technologies and compression technologies would help to assist application opportunities. The potential to improve privacy and security through device based inferences and encrypted knowledge storage would improve user experiences. RiskWise is a scalable platform and will continue to enable research into AI-enhanced contract comprehension.

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Appendix A

Supporting Material

A.1 Sample User Interface Screens

The RiskWise system provides a web-based interface that enables users to upload Terms of Service documents, view analysis results, and interact with the system through a conversational interface. Representative screenshots illustrating major system functionalities are included below.

A.1.1 Document Upload Interface

This screen allows users to select and upload Terms of Service documents for analysis. The interface includes options to choose processing mode and initiate analysis.

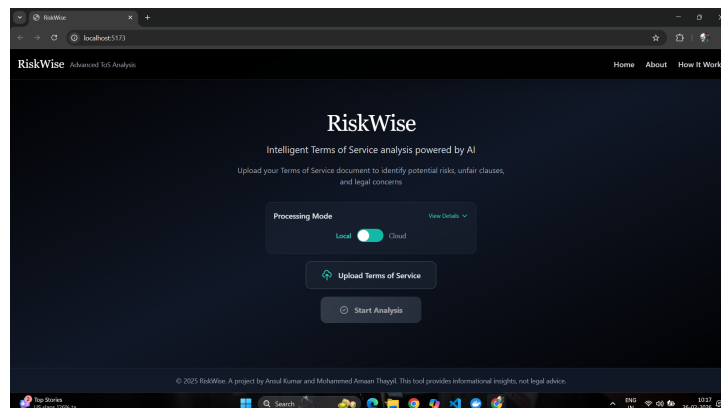


FIGURE A.1: Document Upload Interface - Screen 1

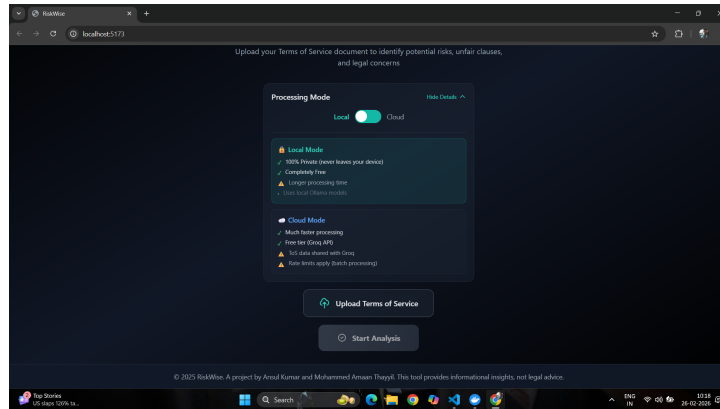


FIGURE A.2: Document Upload Interface - Screen 2

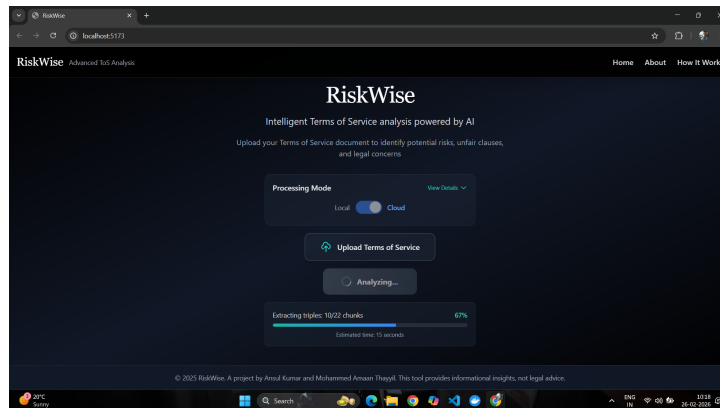


FIGURE A.3: Document Upload Interface - Screen 3

A.1.2 Risk Analysis Results Interface

Following document processing, the system displays categorized risk insights highlighting clauses of potential concern along with severity scores and explanations.

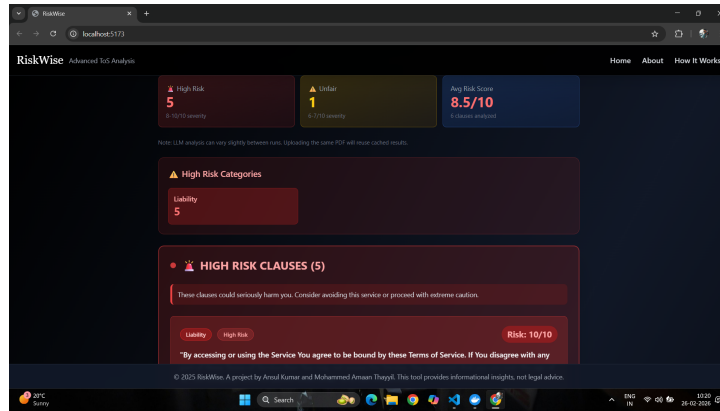


FIGURE A.4: Risk Analysis Results Interface - Screen 1

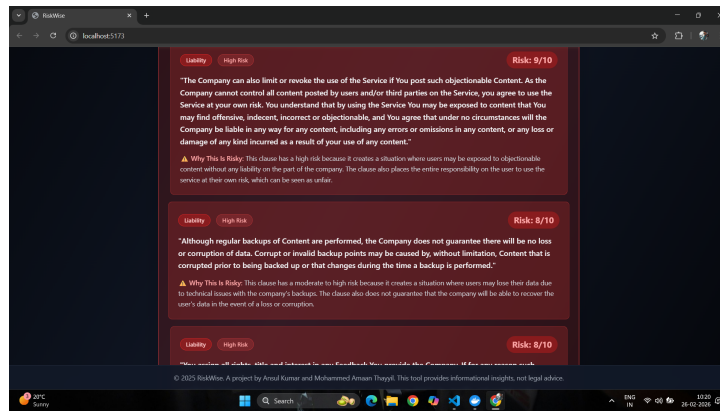


FIGURE A.5: Risk Analysis Results Interface - Screen 2

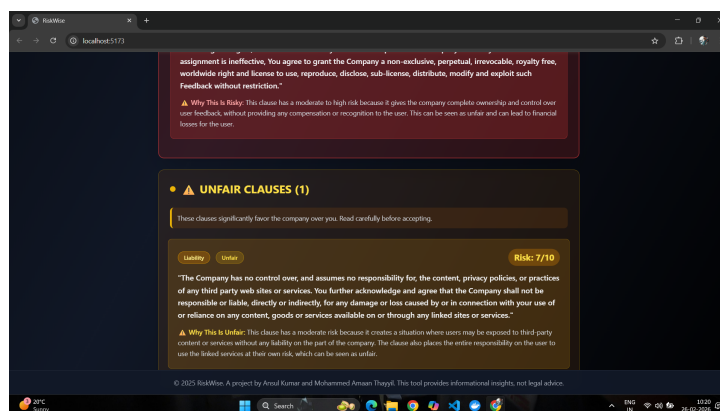


FIGURE A.6: Risk Analysis Results Interface - Screen 3

A.1.3 Conversation Query Interface

The interactive query interface allows users to ask contextual questions regarding identified risks and receive grounded responses based on retrieved document content.

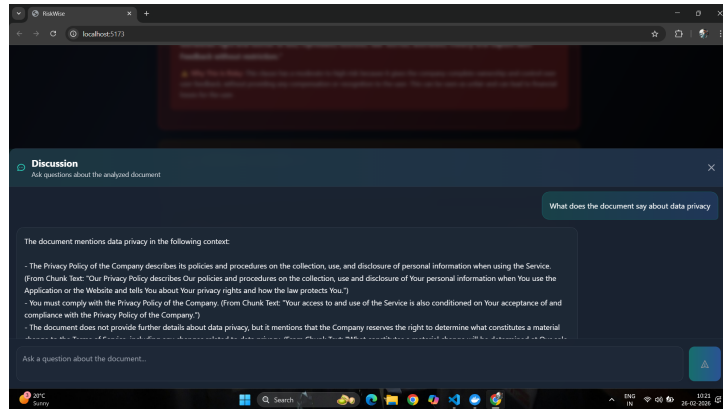


FIGURE A.7: Conversation Query Interface - Screen 1

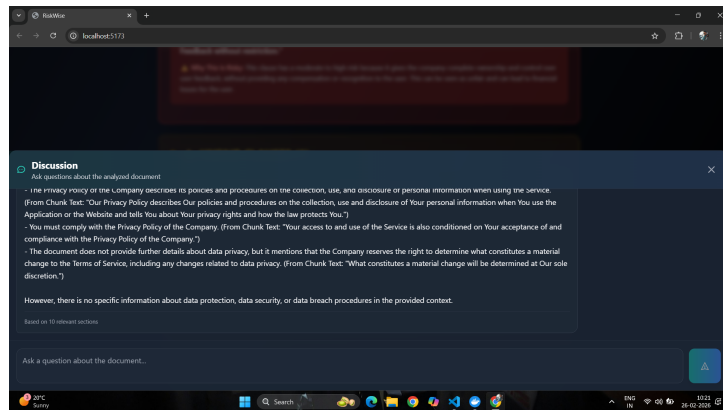


FIGURE A.8: Conversation Query Interface - Screen 2

A.2 Sample Analysis Outputs

This section presents representative outputs generated by the RiskWise system to illustrate risk detection and explanation capabilities.

A.2.1 Detected Risk Clause Example

Example of a clause identified as high-risk, including severity score and explanatory reasoning generated by the system.

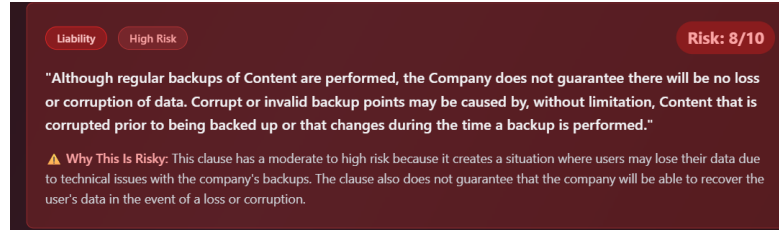


FIGURE A.9: Detected Risk Clause Example

A.2.2 Contextual Question Answering Example

Example illustrating system response to a user query grounded in retrieved document context.

User: What does the document say about data privacy

Chat: The document mentions data privacy in the following context:

- The Privacy Policy of the Company describes its policies and procedures on the collection, use, and disclosure of personal information when using the Service. (From Chunk Text: "Our Privacy Policy describes Our policies and procedures on the collection, use and disclosure of Your personal information when You use the Application or the Website and tells You about Your privacy rights and how the law protects You.")
- You must comply with the Privacy Policy of the Company. (From Chunk Text: "Your access to and use of the Service is also conditioned on Your acceptance of and compliance with the Privacy Policy of the Company.")
- The document does not provide further details about data privacy, but it mentions that the Company reserves the right to determine what constitutes a material change to the Terms of Service, including any changes related to data privacy. (From Chunk Text: "What constitutes a material change will be determined at Our sole discretion.")

However, there is no specific information about data protection, data security, or data breach procedures in the provided context.

Based on 10 relevant sections

A.3 Prompt Design Examples

The system employs carefully designed prompts for knowledge extraction and risk analysis. Representative prompt templates are provided below.

A.3.1 Triple Extraction Prompt

You are an information extraction system specialized in Terms of Service. Your task: From the given text, extract all factual subject–relation–object triples. Rules: - Output ONLY in this exact format, one triple per line: (SUBJECT, RELATION, OBJECT) - No explanations, bullet points, numbering, or extra words. - Keep SUBJECT, RELATION, and OBJECT concise. - If no valid triples exist, return nothing. Examples: Text: "Marie Curie discovered radium and polonium." Output: (Marie Curie, discovered, radium) (Marie Curie, discovered, polonium)

Text: "Users cannot share their passwords with anyone." Output: (User, cannot_share, passwords)

Text: "The User must provide accurate information during account registration." Output: (User, must_provide, accurate information) (User, registers_for, Account)

Text: "The Company may terminate your subscription at any time with notice." Output: (Company, may_terminate, subscription) (Company, gives_notice, User)

Text: "By using the Service, you agree to the Terms of Service and Privacy Policy." Output: (User, agrees_to, Terms of Service) (User, agrees_to, Privacy Policy)

Text: "If the User violates the rules, the Company may suspend the account." Output: (User, violates, rules) (Company, may_suspend, account)

Now extract triples from this text: "(chunk_text)"

A.3.2 Risk Analysis Prompt

Prompt used to instruct the model to identify risky clauses, assign scores, and provide explanations. You are an expert legal document analyst specializing in Terms of Service agreements.

Your task: Identify the 3-5 MOST RISKY clauses that could harm users or limit their rights.

Instructions: 1. Focus on clauses that create LIABILITY, PRIVACY issues, TERMINATION rights, PAYMENT obligations, or UNFAIR CHANGES 2. Ignore routine legal language - find the CONCERNING clauses 3. For EACH risky clause, provide EXACTLY this format with your reasoning:

CLAUSE: [exact text from the ToS - keep it complete] RISK: [number 1-10 where: 1-3=minimal risk, 4-6=moderate risk, 7-10=high risk] CATEGORY: [choose: Privacy, Liability, Termination, Payments, or Changes] REASON: [2-3 sentences explaining specifically why this is risky for users]

IMPORTANT: - Leave ONE blank line between each clause - Risk scores BELOW 7 are NOT truly risky (use 7-10 for actual risks) - Be specific about WHO is harmed and HOW

Terms of Service: (document_text)

Now identify the MOST RISKY clauses. Start with the highest risk items.

Appendix B

Hardware and Software Tools

B.1 Hardware Requirements

The RiskWise system was developed and tested using standard personal computing infrastructure capable of supporting modern web development and machine learning workloads. The hardware configuration used during development is summarized below.

TABLE B.1: Hardware Requirements

Hardware Requirements	
Component	Specification
Processor	Multi-core CPU (Intel i5/i7 or equivalent)
RAM	16 GB system memory
Storage	SSD-based storage for faster indexing and retrieval
GPU (optional)	Not mandatory; used only if local LLM inference required
Network	Stable internet connection for cloud API access

For privacy-focused deployments involving local model execution, higher RAM capacity and GPU acceleration may be beneficial to ensure acceptable inference latency.

B.2 Software Requirements

The RiskWise system employs a full-stack architecture integrating multiple software components across frontend, backend, data storage, and AI processing layers.

TABLE B.2: Software Requirements

Software Requirements		
Category	Software / Tool	Purpose
Operating System	Windows / Linux	Development and execution environment
Programming Language	Python	Backend and AI pipeline implementation
Frontend Framework	ReactJS	User interface development
Backend Framework	FastAPI	API and asynchronous processing
Database	Neo4j	Knowledge graph and vector storage
AI Framework	LangChain	Retrieval orchestration and LLM integration
NLP Libraries	spaCy, Transformers	Text processing and embeddings
Version Control	Git & GitHub	Source code management
Package Manager	pip / npm	Dependency management
Development Environment	VS Code	Code editing and debugging

B.3 Deployment and Runtime Tools

Additional tools supporting system execution and experimentation include:

TABLE B.3: Deployment and Runtime Tools

Deployment and Runtime Tools	
Tool	Purpose
Ollama	Local LLM inference
Groq API	Cloud-based LLM inference
Node.js	Frontend build environment
Browser	User interface access and testing
Mermaid / Diagram tools	System visualization

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